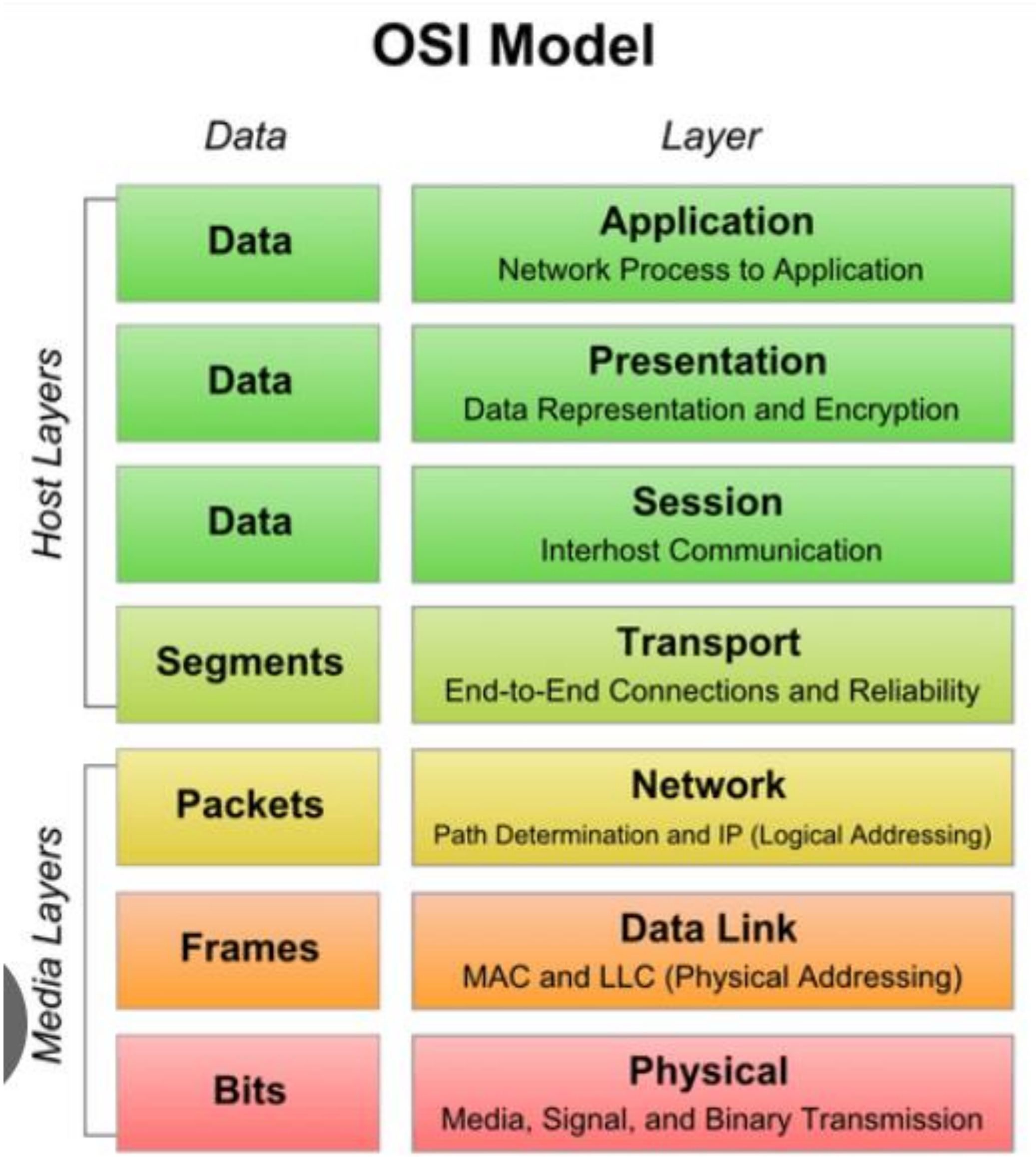
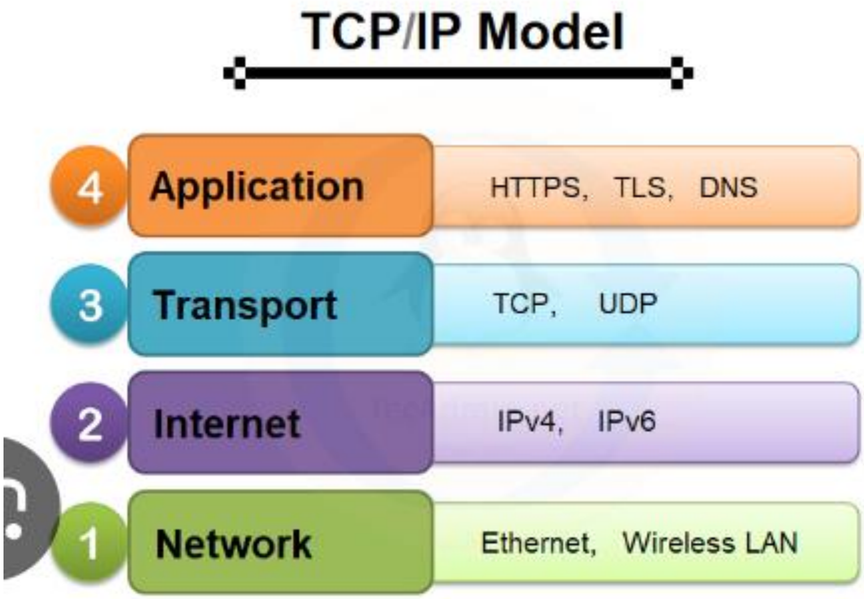
Networking Fundamentals & Hands-on Checks.

1. OSI layers:



L1 Physical layer: Cables, signals, NIC hardware
L2 Data link layer: MAC address, switching.
L3 Network Layer: IP addressing & routing.
L4 Transport layer: TCP/UPD, ports
L5 session layer: session management.
L6 Presentation layer: encryption/formatting.
L7 Application layer: HTTP, DNS,FTP,SMTP

TCP/IP Model :



L1 Network Layer: Sends data inside your local network. Uses MAC addressses. Work with cables, Wi-Fi signals.

L2 Internet layer : This handles IP addressing and routing.
Every device has an IP, Decides where the packet should go, works across different networks.

L3 Transport Layer: Controls Communication between applications.
Uses port numbers: TCP: Reliable used in websites.
UDP: Fast used in streaming.

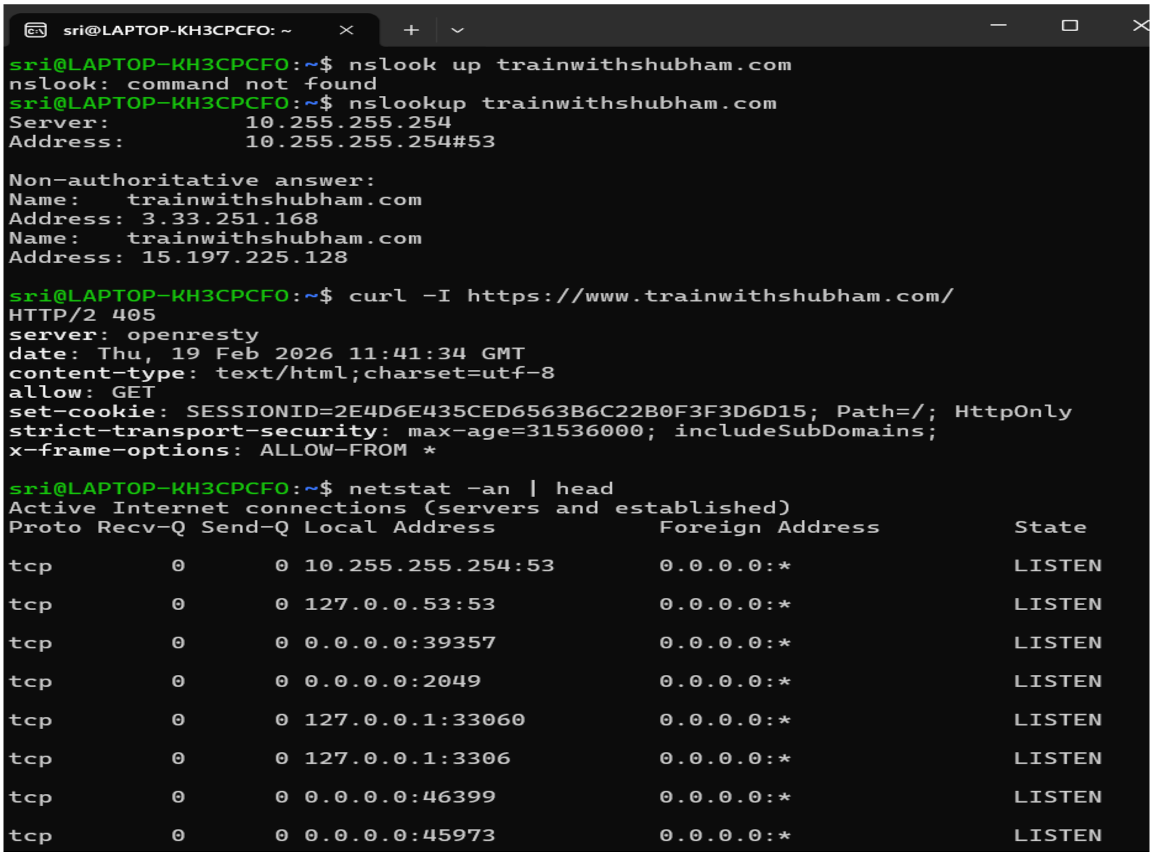
L4 Application layer: What users interact with.

Hands-on Checklist:

1. Identity: hostname –I (192.168.197.173 172.xx.x.x) Shows your machine's IP address.
2. Reachability: Checks if another machine is reachable. (ping <target>)
3. Path: Shows the route your packet takes to reach the target. (traceroute <target> or tracepath)
4. Ports: Shows which services are listening to on your machine. (`ss -tulpn` / `netstat –tulpn)`
5. `Name Resolution:` Converts domain name to IP address. (`dig <domain>` / `nslookup <domain> )`

6. HTTP Check: Sends a quick request to a website and checks response. (curl -I <url>)
7. Connections Snapshot: Shows active network connections. (netstat -an | head)

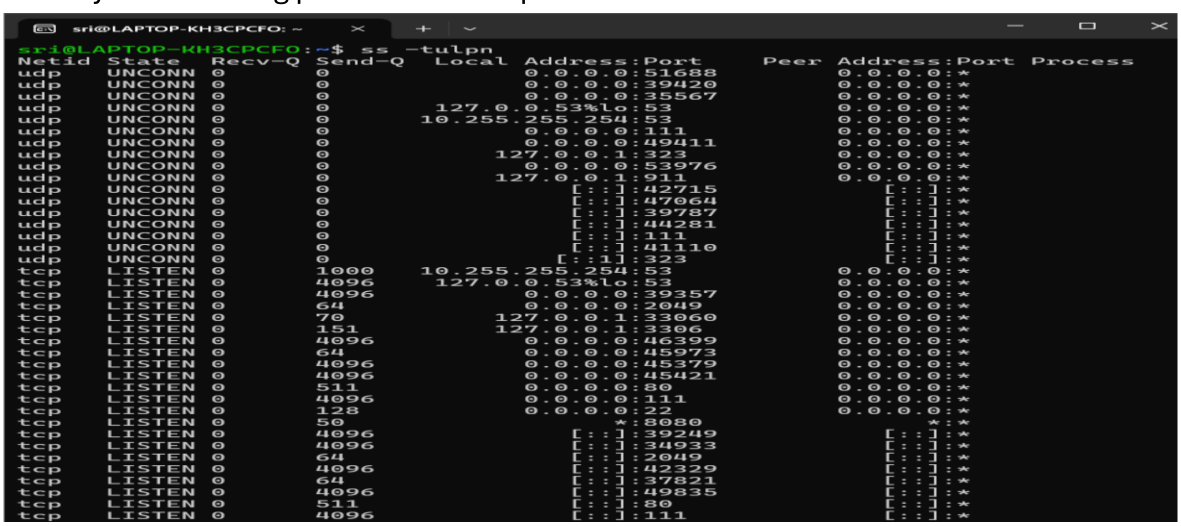
```
sri@LAPTOP-KH3CPCFO: ~
sri@LAPTOP-KH3CPCFO:~$ hostname -I
192.168.197.173 172.17.0.1
sri@LAPTOP-KH3CPCFO:~$ ping trainwithshubham.com
PING trainwithshubham.com (3.33.251.168) 56(84) bytes of data.
64 bytes from aec037177372cc6cd.awsglobalaccelerator.com (3.33.251.168): icm
p_seq=1 ttl=242 time=84.7 ms
64 bytes from aec037177372cc6cd.awsglobalaccelerator.com (3.33.251.168): icm
p_seq=2 ttl=242 time=93.1 ms
64 bytes from aec037177372cc6cd.awsglobalaccelerator.com (3.33.251.168): icm
p_seq=3 ttl=242 time=93.0 ms
64 bytes from aec037177372cc6cd.awsglobalaccelerator.com (3.33.251.168): icm
p_seq=4 ttl=242 time=89.8 ms
64 bytes from aec037177372cc6cd.awsglobalaccelerator.com (3.33.251.168): icm
p_seq=5 ttl=242 time=87.5 ms
64 bytes from aec037177372cc6cd.awsglobalaccelerator.com (3.33.251.168): icm
p_seq=6 ttl=242 time=87.8 ms
^C
--- trainwithshubham.com ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5009ms
rtt min/avg/max/mdev = 84.703/89.328/93.142/3.053 ms
sri@LAPTOP-KH3CPCFO:~$ traceroute trainwithshubham.com
traceroute to trainwithshubham.com (15.197.225.128), 30 hops max, 60 byte pa
ckets
 1  LAPTOP-KH3CPCFO.mshome.net (192.168.192.1)  0.460 ms  0.401 ms  0.374 ms
 2  jiofiber.local.html (192.168.31.1)  96.131 ms  96.109 ms  96.088 ms
 3  192.0.0.1 (192.0.0.1)  88.080 ms  147.354 ms  147.334 ms
 4  192.0.0.1 (192.0.0.1)  147.313 ms  147.291 ms  146.990 ms
 5  192.0.0.1 (192.0.0.1)  146.942 ms  146.850 ms  146.812 ms
 6  192.0.0.1 (192.0.0.1)  148.379 ms  146.546 ms  146.416 ms
 7  192.168.227.99 (192.168.227.99)  146.271 ms  66.385 ms  54.664 ms
 8  172.26.35.52 (172.26.35.52)  48.675 ms 172.26.35.48 (172.26.35.48)  38.3
47 ms  38.683 ms
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  *^C
sri@LAPTOP-KH3CPCFO:~$ ss -tulpn
Netid State  Recv-Q Send-Q  Local Address:Port    Peer Address:Port Process
udp   UNCONN 0      0             0.0.0.0:51688        0.0.0.0:*
```



```
sri@LAPTOP-KH3CPCFO: ~
sri@LAPTOP-KH3CPCFO:~$ nslook up trainwithshubham.com
nslook: command not found
sri@LAPTOP-KH3CPCFO:~$ nslookup trainwithshubham.com
Server:         10.255.255.254
Address:        10.255.255.254#53

Non-authoritative answer:
Name:   trainwithshubham.com
Address: 3.33.251.168
Name:   trainwithshubham.com
Address: 15.197.225.128

sri@LAPTOP-KH3CPCFO:~$ curl -I https://www.trainwithshubham.com/
HTTP/2 405
server: openresty
date: Thu, 19 Feb 2026 11:41:34 GMT
content-type: text/html;charset=utf-8
allow: GET
set-cookie: SESSIONID=2E4D6E435CED6563B6C22B0F3F3D6D15; Path=/; HttpOnly
strict-transport-security: max-age=31536000; includeSubDomains;
x-frame-options: ALLOW-FROM *

sri@LAPTOP-KH3CPCFO:~$ netstat -an | head
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State

tcp        0      0 10.255.255.254:53       0.0.0.0:*               LISTEN

tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN

tcp        0      0 0.0.0.0:39357           0.0.0.0:*               LISTEN

tcp        0      0 0.0.0.0:2049            0.0.0.0:*               LISTEN

tcp        0      0 127.0.0.1:33060         0.0.0.0:*               LISTEN

tcp        0      0 127.0.0.1:3306          0.0.0.0:*               LISTEN

tcp        0      0 0.0.0.0:46399           0.0.0.0:*               LISTEN

tcp        0      0 0.0.0.0:45973           0.0.0.0:*               LISTEN
```

Mini Task: Port probe & Interpret

1. Identify one listening port from ss –tulpn :



```
sri@LAPTOP-KH3CPCFO: ~
sri@LAPTOP-KH3CPCFO:~$ ss -tulpn
Netid State  Recv-Q Send-Q  Local Address:Port   Peer Address:Port Process
udp   UNCONN 0      0             0.0.0.0:51688       0.0.0.0:*
udp   UNCONN 0      0             0.0.0.0:39420       0.0.0.0:*
udp   UNCONN 0      0             0.0.0.0:35567       0.0.0.0:*
udp   UNCONN 0      0       127.0.0.53%lo:53          0.0.0.0:*
udp   UNCONN 0      0      10.255.255.254:53          0.0.0.0:*
udp   UNCONN 0      0             0.0.0.0:111         0.0.0.0:*
udp   UNCONN 0      0             0.0.0.0:49411       0.0.0.0:*
udp   UNCONN 0      0           127.0.0.1:323         0.0.0.0:*
udp   UNCONN 0      0             0.0.0.0:53976       0.0.0.0:*
udp   UNCONN 0      0           127.0.0.1:911         0.0.0.0:*
udp   UNCONN 0      0                [::]:42715          [::]:*
udp   UNCONN 0      0                [::]:47064          [::]:*
udp   UNCONN 0      0                [::]:39787          [::]:*
udp   UNCONN 0      0                [::]:44281          [::]:*
udp   UNCONN 0      0                [::]:111            [::]:*
udp   UNCONN 0      0                [::]:41110          [::]:*
udp   UNCONN 0      0               [::1]:323            [::]:*
tcp   LISTEN 0      1000   10.255.255.254:53          0.0.0.0:*
tcp   LISTEN 0      4096    127.0.0.53%lo:53          0.0.0.0:*
tcp   LISTEN 0      4096          0.0.0.0:39357       0.0.0.0:*
tcp   LISTEN 0      64            0.0.0.0:2049        0.0.0.0:*
tcp   LISTEN 0      70          127.0.0.1:33060       0.0.0.0:*
tcp   LISTEN 0      151         127.0.0.1:3306        0.0.0.0:*
tcp   LISTEN 0      4096          0.0.0.0:46399       0.0.0.0:*
tcp   LISTEN 0      64            0.0.0.0:45973       0.0.0.0:*
tcp   LISTEN 0      4096          0.0.0.0:45379       0.0.0.0:*
tcp   LISTEN 0      4096          0.0.0.0:45421       0.0.0.0:*
tcp   LISTEN 0      511           0.0.0.0:80          0.0.0.0:*
tcp   LISTEN 0      4096          0.0.0.0:111         0.0.0.0:*
tcp   LISTEN 0      128           0.0.0.0:22          0.0.0.0:*
tcp   LISTEN 0      50                  *:8080              *:*
tcp   LISTEN 0      4096             [::]:39249          [::]:*
tcp   LISTEN 0      4096             [::]:34933          [::]:*
tcp   LISTEN 0      64               [::]:2049           [::]:*
tcp   LISTEN 0      4096             [::]:42329          [::]:*
tcp   LISTEN 0      64               [::]:37821          [::]:*
tcp   LISTEN 0      4096             [::]:49835          [::]:*
tcp   LISTEN 0      511              [::]:80             [::]:*
tcp   LISTEN 0      4096             [::]:111            [::]:*
```

2. From the same machine, test it: nc –zv localhost <port> or curl –I hhttp://localhost:<port>

```
sri@LAPTOP-KH3CPCFO:~$ nc -zv localhost 22
Connection to localhost (127.0.0.1) 22 port [tcp/ssh] succeeded!
sri@LAPTOP-KH3CPCFO:~$
```

3. Write one line: is it reachable? If not, what is the next check?

Reachable, the SSH service is listening on port 22 locally; if it wasn't reachable, I would check whether the SSH service is running (`systemctl status ssh`) and verify firewall rules (`ufw status` or `iptables -L`).

1. Which command gives the fastest signal when something is broken?
   => **`ping`** gives the fastest basic signal.

2. What layer would you inspect next if DNS fails? If HTTP 500 shows up?
   = > **If DNS fails:** Inspect the **Application layer** (TCP/IP model). Because DNS operates at:

- Application layer (OSI Layer 7)

3. Two follow-up checks in a real incident:
   1. Check service status : systemctl status <service>
   2. Check logs : journalctl -xe

Networking concepts : DNS, IP,Subnets & Ports

Task 1: DNS – How Names Become IPs

1.Explain in 3-4 lines: What happens when you type google.com in a browser?

=> when you type google.com, your system asks a DNS resolver for its IP address.

The resolver checks cache first; if not found, it queries root -> TLD -> authoritative DNS servers. The final IP address is returned to the browser, which then connects to that IP using HTTP/HTTPS.

2. DNS Record Types :

- A: Maps a domain name to an IPv4 address.

- AAAA: Maps a domain name to an IPv6 address.

-CNAME: Alias of one domain name to another.

-MX: Mail server responsible for receiving emails.

- NS: Specifies authoritative name servers for a domain.

Command : # dig google.com

```
sri@LAPTOP-KH3CPCFO:~$ dig google.com

; <<>> DiG 9.18.39-0ubuntu0.22.04.2-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 16449
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;google.com.                    IN      A

;; ANSWER SECTION:
google.com.             247     IN      A       142.250.195.14

;; Query time: 79 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Thu Feb 19 20:12:44 IST 2026
;; MSG SIZE  rcvd: 55
```

Task 2: IP Addressing:

1. What is an IPv4 address?

=> An IPv4 address is a 32-bit numeric identifier assigned to device on a network. It is written in dotted-decimal format like 192.168.1.10 (4 octets, each 0-255).

2. Public vs Private IP :

Public IP: Routable on the internet.(8.8.8.8)

Private IP: Used inside internal networks.(192.168.1.10)

3. What are the private IP ranges?
    - 10.0.0.0 - 10.255.255.254
    - 172.16.0.0. - 172.31.255.255
    - 192.168.0.0 - 192.168.255.255

```
sri@LAPTOP-KH3CPCFO:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default q
len 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default
 qlen 1000
    link/ether 00:15:5d:78:39:f5 brd ff:ff:ff:ff:ff:ff
    inet 192.168.197.173/20 brd 192.168.207.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe78:39f5/64 scope link
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN g
roup default
    link/ether 02:42:19:a1:0e:4b brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
```

Task 3: CIDR & Subnetting

1.What does /24 mean in 192.168.1.0/24?

- /24 means the first 24 bits are used for the network portion, leaving 8 bits for hosts.
- Usable Hosts
- /24: 254 usable hosts.
- /16: 65,534 usable hosts.
- /28: 14 usable hosts.

2. Why do we subnet?

Subnetting divides large networks into smaller ones to improve performance, security, and IP management.

CIDR	Subnet Mask	Total IPs	Usable Hosts
/24	255.255.255.0	256	254
/16	255.255.0.0	65,536	65,534
/28	255.255.255.240	16	14

Task 4: Ports – The Doors to Services

1. What is a Port?

A Port is a logical communication endpoint that identifies a specific service running on a system.

Why do we need ports?

Ports allow multiple services (web, database,SHH) to run on the same IP without conflict.

Common Ports :

Port	Service
22	SSH
80	HTTP
443	HTTPS
53	DNS
3306	MYSql
6379	Redis
27017	MongoDB

2. Run ss –tulpn match at 2 listening ports to their services?

```
    valid_lft forever preferred_lft forever
sri@LAPTOP-KH3CPCFO:~$  ss -tulpn
Netid  State    Recv-Q  Send-Q    Local Address:Port     Peer Address:Port   Process
udp    UNCONN   0       0               0.0.0.0:51688         0.0.0.0:*
udp    UNCONN   0       0               0.0.0.0:39420         0.0.0.0:*
udp    UNCONN   0       0               0.0.0.0:35567         0.0.0.0:*
udp    UNCONN   0       0         127.0.0.53%lo:53            0.0.0.0:*
udp    UNCONN   0       0        10.255.255.254:53            0.0.0.0:*
udp    UNCONN   0       0               0.0.0.0:111           0.0.0.0:*
udp    UNCONN   0       0               0.0.0.0:49411         0.0.0.0:*
udp    UNCONN   0       0             127.0.0.1:323           0.0.0.0:*
udp    UNCONN   0       0               0.0.0.0:53976         0.0.0.0:*
udp    UNCONN   0       0             127.0.0.1:911           0.0.0.0:*
udp    UNCONN   0       0                  [::]:42715            [::]:*
udp    UNCONN   0       0                  [::]:47064            [::]:*
udp    UNCONN   0       0                  [::]:39787            [::]:*
udp    UNCONN   0       0                  [::]:44281            [::]:*
udp    UNCONN   0       0                  [::]:111              [::]:*
udp    UNCONN   0       0                  [::]:41110            [::]:*
udp    UNCONN   0       0                 [::1]:323              [::]:*
tcp    LISTEN   0       1000     10.255.255.254:53            0.0.0.0:*
tcp    LISTEN   0       4096      127.0.0.53%lo:53            0.0.0.0:*
tcp    LISTEN   0       4096            0.0.0.0:39357         0.0.0.0:*
tcp    LISTEN   0       64              0.0.0.0:2049          0.0.0.0:*
tcp    LISTEN   0       70            127.0.0.1:33060         0.0.0.0:*
tcp    LISTEN   0       151           127.0.0.1:3306          0.0.0.0:*
tcp    LISTEN   0       4096            0.0.0.0:46399         0.0.0.0:*
tcp    LISTEN   0       64              0.0.0.0:45973         0.0.0.0:*
tcp    LISTEN   0       4096            0.0.0.0:45379         0.0.0.0:*
tcp    LISTEN   0       4096            0.0.0.0:45421         0.0.0.0:*
tcp    LISTEN   0       511             0.0.0.0:80            0.0.0.0:*
tcp    LISTEN   0       4096            0.0.0.0:111           0.0.0.0:*
tcp    LISTEN   0       128             0.0.0.0:22            0.0.0.0:*
tcp    LISTEN   0       50                    *:8080                *:*
tcp    LISTEN   0       4096               [::]:39249            [::]:*
tcp    LISTEN   0       4096               [::]:34933            [::]:*
tcp    LISTEN   0       64                 [::]:2049             [::]:*
tcp    LISTEN   0       4096               [::]:42329            [::]:*
tcp    LISTEN   0       64                 [::]:37821            [::]:*
tcp    LISTEN   0       4096               [::]:49835            [::]:*
tcp    LISTEN   0       511                [::]:80               [::]:*
tcp    LISTEN   0       4096               [::]:111              [::]:*
tcp    LISTEN   0       128                [::]:22               [::]:*
sri@LAPTOP-KH3CPCFO:~$
```

Task 5: Putting it Together

1. You run Curl http://myapp.com:8080 - what networking concepts from today are involved?

=> DNS resolves myapp.com to an IP address, then TCP connects to port 8080 on that IP using HTTP protocol.

2. App Can't reach DB at 10.0.1.50.3306 - What to check first?

=> Check network connectivity(ping), verify the database service is running on port 3306, and confirm firewall/security group rules allow access.